

Module 4 — Robot Framework

LAB WORKBOOK

Student Name	Shreya Ghosh
Course	B.Tech — Computer Science and Business Systems (CSBS) (4th Year)
Enrollment No.	12023052018010
Programming Language	Python
Automation Tool	Robot Framework
Academic Year	2026
Module	Module 4 — Robot Framework

This workbook contains the three practical experiments for Module 4: Basic Syntax and Keywords, Variables and Data-Driven Automation, and Custom Keywords and Libraries.

Experiment 1 — Basic Syntax and Keywords

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1. Problem Statement

Create Robot Framework test cases to open a web browser and navigate to a specific URL, use the Input Text keyword to fill form fields, and verify the presence of a specific element using Page Should Contain Element.

2. Objective

To demonstrate basic Robot Framework syntax and browser automation using SeleniumLibrary keywords.

3. Tools, Software and Concepts Used

- Python 3
- Robot Framework
- SeleniumLibrary
- Selenium WebDriver
- Google Chrome and ChromeDriver
- SauceDemo practice application
- Variables and keyword-driven test cases
- Open Browser, Input Text, Page Should Contain Element, Sleep and Close All Browsers

4. Source Code / Implementation

The following implementation was executed using Robot Framework with Python and the specified libraries.

basic_keywords.robot

```
*** Settings ***
Library    SeleniumLibrary

*** Variables ***
${URL}    https://www.saucedemo.com/

*** Test Cases ***
Open Browser And Navigate To Website
    Open Browser    ${URL}    Chrome
    Maximize Browser Window
    Sleep    2s
    [Teardown]    Close All Browsers

Fill Login Form
    Open Browser    ${URL}    Chrome
    Maximize Browser Window
    Input Text    id=user-name    standard_user
    Input Text    id=password    secret_sauce
```

```
Sleep      2s
[Teardown]  Close All Browsers

Verify Element On Webpage
  Open Browser    ${URL}    Chrome
  Maximize Browser Window
  Page Should Contain Element    id=login-button
  Sleep      2s
  [Teardown]  Close All Browsers
```

5. Output

The browser was opened and navigated to the SauceDemo URL. The login form fields were populated successfully, and the login button was verified as present on the webpage.

6. Result

The three Robot Framework test cases executed successfully with 3 passed and 0 failed.

7. Observation

The test cases demonstrate the readable keyword-driven syntax of Robot Framework and the use of SeleniumLibrary for basic browser interactions.

8. Conclusion

The basic Robot Framework syntax and required SeleniumLibrary keywords were successfully implemented for browser navigation, form input, and element verification.

Experiment 2 — Variables and Data-Driven Automation

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1. Problem Statement

Implement a Robot Framework test case using variables for usernames and passwords and perform the same login actions on multiple data sets supplied through an external CSV file.

2. Objective

To demonstrate the use of variables, a test template, and external CSV data for repeated Robot Framework automation.

3. Tools, Software and Concepts Used

- Python 3
- Robot Framework
- SeleniumLibrary
- DataDriver
- Selenium WebDriver
- Google Chrome and ChromeDriver
- CSV test data
- Variables, Test Template and data-driven testing

4. Source Code / Implementation

The following implementation was executed using Robot Framework with Python and the specified libraries.

data_driven_login.robot

```
*** Settings ***
Library      SeleniumLibrary
Library      DataDriver      file=login_data.csv

Test Template  Login With Different Users
Test Teardown  Close All Browsers

*** Variables ***
${URL}         https://www.saucedemo.com/

*** Test Cases ***
Login Test     Default User      Default Password

*** Keywords ***
Login With Different Users
    [Arguments]  ${username}  ${password}
    Open Browser  ${URL}      Chrome
    Maximize Browser Window
```

```
Input Text    id=user-name    ${username}  
Input Text    id=password    ${password}  
Click Button  id=login-button  
Page Should Contain Element  class=title  
Sleep        2s
```

5. Output

The data-driven login implementation uses the external CSV file to supply multiple username and password combinations. The same login flow is applied to each supplied data set.

6. Result

The data-driven automation was implemented using Robot Framework DataDriver and an external CSV file.

7. Observation

Separating the test data from the test flow allows the same automation steps to be reused for different input values.

8. Conclusion

Variables and data-driven automation were successfully demonstrated using Robot Framework and external CSV test data.

External test data — login_data.csv

```
*** Test Cases ***;${username};${password}  
Login with standard user;standard_user;secret_sauce  
Login with problem user;problem_user;secret_sauce
```

Experiment 3 — Custom Keywords and Libraries

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1. Problem Statement

Build a custom Robot Framework keyword using Python to perform a specific action such as adding two numbers, and utilize the BuiltIn library to perform a mathematical operation.

2. Objective

To create a custom Python keyword and demonstrate the use of Robot Framework's BuiltIn library for mathematical operations and result validation.

3. Tools, Software and Concepts Used

- Python 3
- Robot Framework
- BuiltIn library
- Custom Python library
- Custom keywords
- Evaluate
- Should Be Equal As Numbers
- Visual Studio Code
- Python virtual environment

4. Source Code / Implementation

The following implementation was executed using Robot Framework with Python and the specified libraries.

custom_library.py

```
def calculate_sum(a, b):
    return int(a) + int(b)
```

5. Output

The custom Python keyword Calculate Sum successfully calculated $10 + 20$ and returned 30. The BuiltIn Evaluate keyword calculated 15×4 and returned 60. Both results were validated.

6. Result

The two Robot Framework tests executed successfully with 2 passed and 0 failed.

7. Observation

The Python library extends Robot Framework with a user-defined keyword, while BuiltIn provides general-purpose keywords for evaluation and validation.

8. Conclusion

The custom Python keyword was successfully integrated with Robot Framework, and the BuiltIn library was successfully used for mathematical evaluation and verification.

Robot Framework test file — custom_keywords.robot

```
*** Settings ***
Library      BuiltIn
Library      custom_library.py

*** Test Cases ***
Custom Python Keyword Test
    ${result}=    Calculate Sum    10    20
    Should Be Equal As Numbers    ${result}    30

BuiltIn Mathematical Operation Test
    ${result}=    Evaluate    15 * 4
    Should Be Equal As Numbers    ${result}    60
```